Xinchen Gao

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EDUCATION

Columbia University (Columbia Business School & Fu Foundation School of Engineering) New York, NY
Major: M.S. in Business Analytics (STEM) *Expected 12/2026*
Core Courses: Machine Learning, Statistical Modeling, Optimization, Analytics in Practice (clinical capstone).

Capital University of Economics and Business Beijing, China
Major: B.M. in Information Management and Information Systems *09/2021-07/2025*

RESEARCH INTERESTS

Clinical risk prediction, bias and reliability of LLMs in medicine, and real-world clinical data (registries, EHRs).

RESEARCH EXPERIENCE

Clinical AI for Surgical Risk Stratification in Neonatal Congenital Heart Disease New York, NY
Research Capstone *at Columbia University Irving Medical Center (Nimrod Goldshtrom, MD, MS)* *01/2026-05/2026*

- **Center-specific surgical risk model (mortality and length of stay):** Identified that national registry scores (STS, PC4), fit on broad pediatric populations, are least reliable for the high-risk neonates that drive surgical decisions. Formulated in-hospital mortality and length-of-stay prediction on a **20-year single-center cohort of 1,500+ neonatal cardiac surgeries** (100+ variables) and built a reproducible REDCap pipeline that **cut case preparation by ~30%**. Adopted **temporal validation** to reflect two decades of practice drift, and found that the apparent gain of gradient boosting over penalized logistic regression **did not hold across eras** given the low event count; therefore retained the **interpretable model as primary**, used XGBoost with SHAP only to check for missed drivers, and evaluated by **calibration and decision curve analysis** rather than AUROC alone.
- **Multi-LLM chart abstraction (correcting single-model bias):** Used LLMs to map de-identified free text and inconsistently coded fields (anatomic subtype, prior interventions, complications) to STS codes. Against a clinician-abstracted subset, **identified systematic bias in a single LLM**: rare anatomies were pulled toward common diagnoses, and outputs changed when the same note was reordered. Proposed a **cross-family multi-LLM consensus** in which each model must quote supporting evidence and is given retrieved STS code definitions, with disagreements routed to clinician review. Found that majority voting still failed on the rarest categories because **errors were correlated across models**, and added a rarity-aware rule that sends low-frequency codes to review regardless of agreement, concentrating clinician effort on genuinely ambiguous cases.
- **Uncertainty-aware risk reporting:** Designed patient-level outputs that pair each risk with a **conformal prediction interval**, its top contributing factors, and the **number of comparable historical cases** from nearest-neighbor matching, so that wide intervals on rare anatomies read as an explicit extrapolation warning rather than a defect. As client lead with neonatologists, cardiac intensivists, and surgeons, integrated the tool into the hospital's weekly multidisciplinary case conference.

Evaluating and Improving LLM Agents for Automated Analytical Reporting Beijing, China
AI Analytics Intern *at Meituan* *05/2026-08/2026*

- **Stage-level evaluation of LLM report agents:** Identified that end-to-end checks on agent-generated weekly and monthly reports could not tell which step had failed. Designed an evaluation framework that scores each stage separately (SQL correctness, extraction completeness, cross-source stitching mismatch, metric consistency), combining rule-based checks, **LLM-as-a-judge**, and human review. Failure tracing showed that many recurring errors came from **inconsistent metric definitions across data sources** rather than from SQL generation; introduced a shared metric dictionary retrieved by the agent before querying, **reducing recurring report errors by 24.5%**.
- **Reliability of the LLM judge:** Calibrating the LLM judge against human-labeled reports revealed that it **rewarded fluent, confident prose even when the underlying numbers were wrong**. Decoupled numeric verification from writing quality, so that figures are checked programmatically against source tables and the LLM judge scores only structure and interpretation against a fixed rubric, and kept a rotating human audit sample to detect judge drift.
- **Agent skills and anomaly detection:** Packaged stable workflows (SQL generation, data retrieval, Excel stitching, report

synthesis) into **reusable agent skills** adopted across teams, **saving ~10 hours** of manual analysis per week. Built an AI-assisted dashboard for brand-level anomaly detection; found that shifts in segment mix produced false alarms in weighted metrics, and separated **rate effects from mix effects** so that alerts reflect real performance changes and speed up root-cause analysis.

Domain-Adapted Language Models for Retail Investor Sentiment Beijing, China

Research Assistant *at Capital University of Economics and Business* *03/2024-07/2024*

- **Domain adaptation for informal financial text:** Observed that general sentiment models systematically misread sarcasm and platform slang in retail investor posts, biasing any downstream sentiment measure. Built a **concurrent, multithreaded scraper** (Selenium, BeautifulSoup) to collect one year of comments on multiple stocks from Eastmoney Guba, one of China's largest retail investor forums. **Fine-tuned FinBERT** and benchmarked it against the off-the-shelf model and **zero-shot LLM classification**. Traced the remaining errors to label noise in short, ambiguous posts, revised the annotation guideline through disagreement review, and retrained; the adapted model was **more accurate and more consistent on slang and sarcasm**, and was used to build stock-level sentiment series to study how retail mood moves with market activity.

TEACHING EXPERIENCE

Columbia University, Department of IEOR New York, NY

Course Assistant*, IEOR E4735 Structured & Hybrid Products* *10/2026-Present*

- Support graduate students in quantitative pricing and risk analysis of structured products through office hours, assignment grading, and review of modeling exercises.

INTERNSHIP

Tencent Beijing, China

Data Analysis Intern *11/2024-12/2024*

- **Funnel and churn analysis:** Modeled viewing funnels for **12.1M Tencent Video users** with SQL window functions (RANK, LEAD), tracing journeys from app open to playback, completion, and next-video start, and localized the largest drop-off to the moment after the first video ends. Comparing retained and churned users on watch time, session frequency, and recommendation clicks in Python pointed to the recommendation shown at that point as the likely cause.
- **Randomized experiment:** Designed a **randomized A/B test** of the recommendation flow with a pre-specified primary metric, guardrail metrics, and a power analysis. Tracked the daily treatment effect to rule out a short-lived novelty effect before reading out the result; the redesign **increased next-video play rate by 4.5%**.

Ernst & Young (EY) Beijing, China

Integrity & Compliance Intern *07/2023-08/2023*

- Replaced manual reconciliation of physician engagement and reimbursement records with a **Python pipeline using rule-based validation and anomaly flagging**, **saving ~5 hours** of processing per week, and validated **150+ records per day** with SQL and Excel reconciliation, resolving mismatches and missing fields across activity logs, event records, and payment documents for audit-ready reporting.

Beijing Xinshichuangyu Technology Co., Ltd. Beijing, China

Data Analysis Intern *01/2024-03/2024*

- Sessionized **2GB+ of gameplay logs** in SQL and Python and reconstructed entry, match start, and completion funnels to quantify step-level conversion loss; built standardized D1 retention and participation datasets from **300K+ MySQL player records** with index-backed queries, then designed and validated an **A/B test** on event entry flow and reward visibility that **raised player participation by 18%**.

SKILLS

- Statistics & ML: **penalized regression**, **calibration and decision curve analysis**, **conformal prediction**, missing-data imputation, gradient boosting (XGBoost), SHAP, time series forecasting
- Experimental Design: **randomized A/B testing**, power analysis, guardrail metrics, experiment monitoring
- Clinical Data: **REDCap**, de-identified registry data, diagnosis and procedure code encoding, temporal validation
- LLM & NLP: **LLM bias evaluation**, **multi-LLM consensus**, **LLM agent evaluation**, LLM-as-a-judge, transformer fine-tuning (Hugging Face), RAG, function calling, MCP
- Proficient in **Python** (PyTorch, scikit-learn, statsmodels, pandas), **R**, **SQL**; Git, AWS, Tableau